ROCKS & MINERALS KEY



Staten Island Technical High School
CSE 2025

STATION 1: Fluorite, Halite (11 pts)

1. Yes (1); glows blue-violet (2) under ultraviolet light (2)
2. B (2)
3. Transparent to translucent (2)
4. Penetration twinning (2)

STATION 2: Sphalerite, Galena, Stibnite (10 pts)

5. 3.91 (2)
6. Subconchoidal (2)
7. Any of the following are acceptable: sedimentary limestone deposits, hypothermal veins, mesothermal veins, hydrothermal replacement deposits, basalts (2)
8. Blende (2)
9. B & C (2; no points for only one)

STATION 3: Obsidian, Basalt, Scoria (10 pts)

10. Smooth, glassy (also accept vitreous and hyaline) (2)
11. True (2)
12. Gabbro (2)
13. If molten rock solidifies before the gas escapes, the bubbles become small rounded or elongated cavities (2)
14. It is not crystalline, which is a defining characteristic of minerals (2)

STATION 4: Shale, Bituminous Coal (10 pts)

15. True (2)
16. Slate (2)
17. Black (2)
18. Fissility (2)
19. 2-2.5 (2)

STATION 5: Arkose, Quartz sandstone (10 pts)

20. A - Arkose (1); B - Quartz sandstone (1)
21. Conchoidal (2)
22. False; This specimen is highly porous (2)
23. Alexandre Brongiart (2) 1826 (2)

STATION 6: Serpentinite, Mica schist (10 pts)

24. 2.73-3.19 g/cm^3 (2)
25. Shiny (2)
26. Accept any two of the following: decoration, building or paving stone, natural cement, art, jewelry, artifacts, monuments (2)
27. Chrysotile asbestos (2); mesothelioma (2)

STATION 7: Dolomite, Azurite, Rhodochrosite (10 pts)

28. $CaMg(CO_3)_2$ (2)
29. C (2)
30. Earthy smell (2)
31. 1.720 - 1.848 (2)
32. Containing no water (2)

STATION 8: Sulfur, Gold (12 pts)

33. It melts into a blood-red liquid (2) and emits a blue flame (2)
34. Smell and taste like rotten eggs (2)
35. Pyrite (2)
36. Accept either insecticide/fungicide OR nutrient for plants (2)
37. Karats (2)

STATION 9: Conglomerate, Bauxite (10 pts)

38. Comes from the Latin *conglomeratus* (2) meaning “to roll together” (2)
39. Aluminum (2)
40. Strong water current produced the rounded shape on the pebbles (2)
41. Yes (2)

STATION 10: Calcite, Malachite (10 pts)

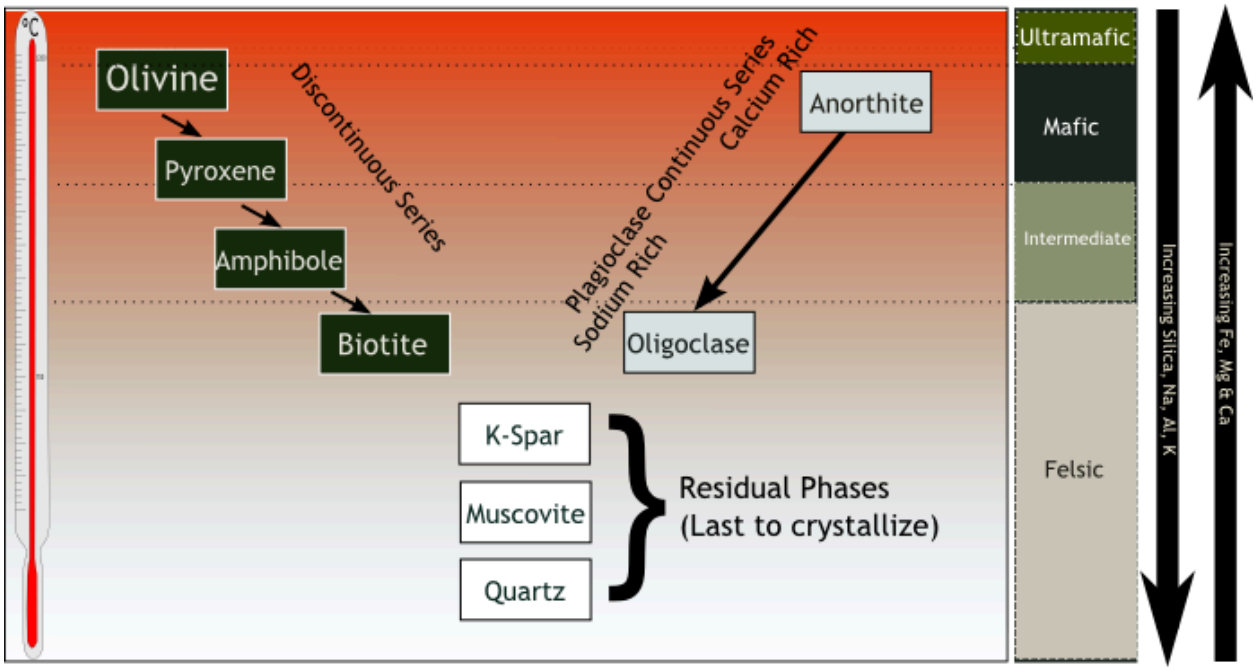
42. Retrograde solubility (2)
43. Azurite (2)
44. Brittle (2)
45. Chalk (2)
46. Trigonal (2)

STATION 11: Breccia, Greywacke (12 pts)

47. Accept either of the following: outcrop bases where mechanical weathering debris accumulates OR stream deposits (2)
48. 6-7 (2)
49. The goddess Taweret (2)
50. No (2)
51. Megabreccia (2); form from landslides, impact events, or caldera collapse (2)

STATION 12: Bowen’s Reaction Series (16 pts)

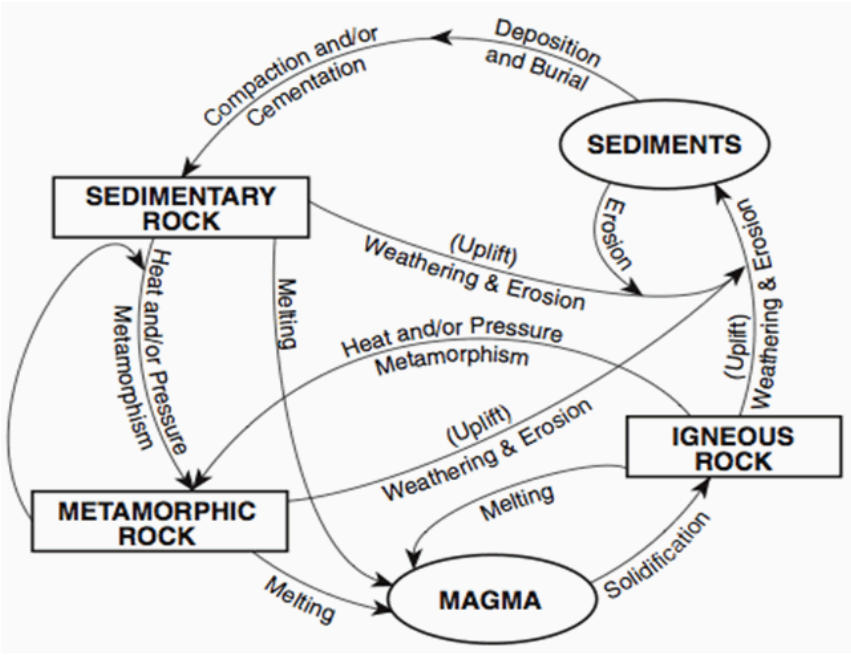
52. Proper sketch of discontinuous series (2); plagioclase continuous series (2); residual phases (2); and the rock type and temperature axes (2). See the below image for reference:



53. The discontinuous branch involves minerals that form to produce new and different minerals (1), while the continuous branch involves an ongoing change in the composition of a single mineral (1)
54. 1920s and 30s (2)
55. Certain minerals crystallize at higher temperatures because they have simpler crystal structures and are more stable at high temperatures (2), while others form at lower temperatures because they have more complex structures and require cooler conditions to become stable (2)

STATION 13: The Rock Cycle (13 pts)

56. 1 pt for each term (5). See the filled in chart:
57. Rocks buried deep within the Earth's crust are brought to the surface (2)
58. Transforms existing rocks into magma and then back into new igneous rocks (2)
59. Transforms loose sediments into solid sedimentary rocks (2)
60. Erosion is the process of moving sediment from one place to another, while deposition is the settling of the sediment (2)



STATION 14: Igneous Structures (14 pts)

61. E (1); intrusive (1)
62. A (1); intrusive (1)
63. F (1); extrusive (1)
64. B (1); intrusive (1)
65. D (1); intrusive (1)
66. G (1); extrusive (1)
67. C (1); intrusive (1)

STATION 15: Environments of Deposition (12 pts)

68. Alluvial fan (2)
69. Lacustrine deposit (2)
70. Delta (2)
71. Aeolian (2)
72. Alluvial, continental (1); aeolian, continental (1); lacustrine, continental (1); delta, transitional (1)

STATION 16: Relative Dating (15 pts)

73. E, G, B, D, A, F, C (1 pt for each letter in the right order) (7; +2 if all are in correct order)
74. Law of Superposition (2)
75. Law of Cross-cutting Relationships (2)
76. Law of Cross-cutting Relationships(2)

STATION 17: Rapid ID (10 pts)

77. Kyanite (1)
78. Peridotite (1)
79. Tuff (1)
80. Travertine (1)
81. Magnetite (1)
82. Chalcopyrite (1)
83. Agate (1)
84. Pumice (1)
85. Gneiss (1)
86. Satin spar (1)

CSE - Repaired practice edition - Answer key

Division C | Rocks And Minerals | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

9. (2 points) Authored replacement for source item 9: The repair supplies three specimens: quartz (piezoelectric), galena (not piezoelectric), and stibnite (not piezoelectric). Which specimen can generate electric polarization under suitable mechanical stress?

answer: Quartz. The relevant property is piezoelectricity; do not use the inconsistent B/C pair in the source key. (2 points).

34. (2 points) Authored replacement for source item 34: A clean elemental sulfur specimen is odorless at room temperature. Explain why a rotten-egg odor would not, by itself, identify that specimen as sulfur.

answer: Rotten-egg odor is associated with hydrogen sulfide, a different substance. Do not equate a compound containing sulfur with pure elemental sulfur or use smell/taste as a definitive sample identification. (2 points).

63. (2 points) Authored replacement for source item 63: A volcanic neck consists of magma solidified inside a volcanic conduit and exposed later by erosion. Is the solidified conduit an intrusive or extrusive body?

answer: Intrusive (hypabyssal/shallow intrusive); it solidified inside the conduit rather than as a surface lava flow. (2 points).